N149AV (S/N: KONUK10295942) — Connectivity Diagnostic Report

Immfly Ground Operations / IT — Session: August 21, 2026

Field	Value
Machine	N149AV (S/N: KONUK10295942)
Affected component	Sierra Wireless EM7565 modem / PPP session (cellular connectivity)
Initial symptom	Loss of internet connectivity reported by user
Modem radio status	Healthy — registered, 97-100% signal, after several recovery cycles
Final connectivity status	UNRESOLVED — modem registered on cellular network but no real IP traffic
Confirmed root cause	Not determined with certainty; AT ports (ttyUSB3/4/5) stopped responding to AT commands after multiple disable/enable cycles
Recommended action	Escalate for physical inspection / possible modem replacement; monitor for recurring USB disconnection pattern

Narrative Summary

During a work session focused on another machine (N098AV), the user reported that N149AV — a unit previously repaired and confirmed functional in an earlier session (where the robust **modem-enable-check.timer** modem-restart pattern was installed) — had lost internet connectivity again. This was investigated in parallel while N098AV completed an independent image capture.

Initial diagnostics showed the modem itself was physically absent from the USB bus at the time of the first check (`mmcli -L` found no modem). Reviewing `dmesg` confirmed a real, spontaneous physical USB disconnection followed by an automatic reconnection about 9 seconds later — an isolated event, not a recurring disconnection pattern (only 1 occurrence recorded in the kernel buffer).

From there, a progressive recovery sequence was executed: restarting the ModemManager service, manually enabling the modem, restarting the PPP session, a full radio-level disable/enable cycle, and finally a full machine reboot. At each stage the modem returned to a *registered* state with high signal (97-100%), and on several occasions the PPP session successfully negotiated (LCP, PAP authentication, IPCP, IP assignment and peer IP) — but real traffic (ping) never got through, not even to the immediate PPP peer (10.64.64.64), ruling out an upstream carrier routing issue and pointing to a more local problem in the data session negotiation.

An accumulation of up to 16 zombie `pppd` processes running simultaneously after the machine reboot was additionally detected, all competing for the same locked port (`/dev/modemPPP`), a symptom of the automatic relaunch cron job (`/etc/cron.d/ansible`) continuing to spawn new attempts every minute without prior ones terminating cleanly. After clearing all processes and forcing a manual foreground attempt, the exact failure point was identified: the connect script (`chat`) consistently failed at the "Initializing modem" step with "Connect script failed", indicating the AT port was not responding to commands, whether due to an access conflict with

ModemManager or a deeper issue with the modem itself.

A possible port-mapping conflict was investigated (recalling that N149AV, due to its older firmware, uses a different USB port layout than the rest of the fleet: AT on ttyUSB5 instead of ttyUSB4). After freeing the port from ModemManager (confirmed free via `fuser`) and testing direct AT communication on all three candidate serial ports (ttyUSB3, ttyUSB4, ttyUSB5) using both minicom and direct device write/read (`stty` + `tee`/`cat`), **none of the three ports responded to the AT command**, even though the modem continued to be correctly detected at the USB level (`lsusb`) and by ModemManager via QMI. This is a notable finding: the QMI channel (used for radio registration and signal) continues to work, but the classic AT channel (required for the PPP negotiation this specific machine uses) does not respond at all.

Given the time already invested in diagnostics and the need to continue other tasks in parallel (image capture on N098AV), it was decided to pause active troubleshooting and document the current state for later intervention or escalation.

Troubleshooting Steps Performed

1. Initial diagnostics

mmcli -L → 'No modems were found'. mmcli -m 0 → 'couldn't find modem'. Confirmed total absence of the modem at the time of the first check.

2. Auto-recovery timer review

modem-enable-check.timer confirmed active and running every 2 minutes without errors (Succeeded on every cycle), but several cycles showed 'No modem detected, skipping' — confirms the timer is working correctly, but cannot act on an absent device.

3. dmesg analysis

Confirmed 1 'USB disconnect' event followed by automatic reconnection ~9 seconds later. Only 1 occurrence in the full buffer — an isolated event, not a repeating pattern so far.

4. ModemManager restart

sudo systemctl restart ModemManager. The modem reappeared as modem0 in 'disabled' state. Manually enabled with mmcli -m 0 --enable → registered, 100% signal.

5. PPP verification

ppp0 existed with an assigned IP, but ping -I ppp0 8.8.8.8 → 100% packet loss. Review of the pppd log showed successful, complete LCP/PAP/IPCP negotiation — the session looked healthy at the protocol level but had no real traffic.

6. PPP session restart

The active pppd process was killed and the automatic cron relaunch was awaited. New PPP session with a new IP, same successful negotiation, same result: no traffic even to the immediate PPP peer (10.64.64.64).

7. Full radio cycle (disable/enable)

mmcli -m 0 --disable followed by --enable, forcing a full re-registration on the cellular network. Result: registered, 100% signal — but real IP connectivity still did not work after relaunching PPP again.

8. Full machine reboot

sudo reboot. After boot: modem registered (97% signal), but ppp0 did not exist yet. 16 accumulated, simultaneously running pppd processes were detected, spawned every minute by the cron job with none completing or terminating — all locked, competing for the same port.

9. Process cleanup and chat script diagnosis

sudo pkill -9 pppd removed the 16 zombie processes. pppd was run manually in the foreground (nodetach) to observe the real error: the connect script (chat) consistently failed at 'Initializing modem' with 'Connect script failed', unable to communicate with the modem via AT.

10. Port conflict verification

fuser confirmed ModemManager had /dev/ttyUSB5 open. The modem was disabled (mmcli -m 0 --disable) and the port was confirmed free (empty fuser output). Despite this, the chat script kept failing at the same point.

11. Direct AT test on all three candidate ports

Direct AT communication was attempted (via minicom and via stty + tee/cat) on ttyUSB3, ttyUSB4, and ttyUSB5 — the three serial ports exposed by the modem. None responded to the AT command, even though lsusb continued to confirm the modem's physical presence on the bus and ModemManager continued to see it correctly via QMI (registered, high signal).

Conclusion and Recommendation

N149AV's modem maintains functional radio registration (QMI channel operational, 97-100% signal), but the classic AT command channel — required for the PPP negotiation this specific machine needs due to its older firmware — stopped responding on all three candidate serial ports following a spontaneous physical USB disconnection and the multiple recovery cycles performed during this session. This suggests a possible partial or intermittent modem failure at the firmware/hardware level, different in nature from the total SIM failure cases (N096AV, N010AV) or the complete modem absence from the bus (N120AV), but which could represent an early stage of similar degradation.

Recommended next steps: (1) Update N149AV's modem firmware to match the rest of the fleet's standard (SWI9X50C_01.14.02.00) using the already-established procedure — this would also resolve the root issue of this machine depending on the chat/AT mechanism instead of pure QMI for its data connection, aligning it with the rest of the fleet. (2) Keep N149AV under observation for possible recurring physical USB disconnections over the next few days. (3) If the problem persists after the firmware update, escalate to Luisa for physical modem replacement evaluation, following the same workflow used for N096AV, N010AV, and N120AV.